MIL-I-5416B
15 JANUARY 1962
SUPERSEDING
MIL-I-5416A
17 AUGUST 1960

MILITARY SPECIFICATION

INDICATOR, INDICATED AIRSPEED, 10-150 KNOTS

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force

1. SCOPE

1.1 This specification covers one type of pitot static airspeed indicator, range 10 to 50 knots

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein

SPECIFICATIONS

FEDERAL

PPP-B-636 — Box, Fiberboard

MILITARY

MIL-P-116 — Preservation, Methods of.

MIL-I-5415 — Indicators, Airspeed, Pitot and Static Pressure, General Specification for

MIL-P-7936 — Parts and Equipment, Aeronautical, Preparation for Delivery

MIL-L-25142 — Luminescent Material, Fluorescent.

STANDARDS

FEDERAL

FED STD No 595 — Colors

MILITARY

MIL-STD-129 — Marking for Shipment and Storage.

MS28045 — Indicator, Indicated Airspeed, Pitot Static 10-150 Knots

MS33585 — Pointers, Dial, Standard Design of Aircraft Instrument.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Qualification.** The indicator furnished under this specification shall be a product, samples of which have been tested and passed the qualification tests specified herein, and

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have been listed on or approved for listing on the applicable Qualified Products List

3.2 General. The general requirements specified in Specification MIL-I-5415 are applicable as general requirements of this specification Where the requirements of Specification MIL-I-5415 and this specification conflict, the requirements of this specification shall govern

3.3 Material. Material shall be in accordance with the requirements specified in Specification MIL-I-5415

3.4 Design and construction.

3.4.1 *Range.* The airspeed indicator shall be designed to satisfactorily operate in the range of 10 to 150 knots.

3.4.2 *Dimensions.* The airspeed indicator shall conform to the dimensions shown on Standard MS28045

3.4.2.1 *Case finish.* The case shall be finished with a lusterless black material, Color No 37038 of Federal Standard No 595 The finishing material shall be of a durable type to withstand usage encountered in service

3.4.3 *Pointer.* The pointer shall conform to MS33585-3 The shaded portion of the pointer shall be finished in fluorescent material conforming to Specification MIL-L-25142 for Air Force procurement and in lusterless white paint conforming to Color No 37875 of Federal Standard No 595 for Navy procurement The unshaded portion of the pointer shall be finished in lusterless black, Color No 37038 of Federal Standard No 595

3 4.3 1 *Limitation of pointer movement* The pointer movement shall be limited by suitable stops in the mechanism in such a way that the pointer will not be permitted to rotate more than 350 degrees of arc clockwise from the zero graduation when the indicator is submitted to over pressure tests Suitable stops may also be incorporated in the indicator mechanism to limit the counterclockwise rotation of the pointer to the zero graduation. The design of the stops shall be subject to the approval of the procuring activity

3.5 Weight. The weight of each indicator shall not exceed 1 pound

3.6 Performance. The airspeed indicators shall pass the following tests specified in section 4 of Specification MIL-I-5415

Damping	Leakage
Low temperature exposure	Scale error at room temperature
High temperature exposure	Friction
Acceleration	Weight
Humidity	Magnetic effect
Salt spray	Position error
Aging	Pointer zero position error
Vibration failure	Low temperature operation
Mounting nuts	High temperature operation
Mounting lugs	Vibration error
Fungus resistance	Overpressure
Examination of product	Seasoning

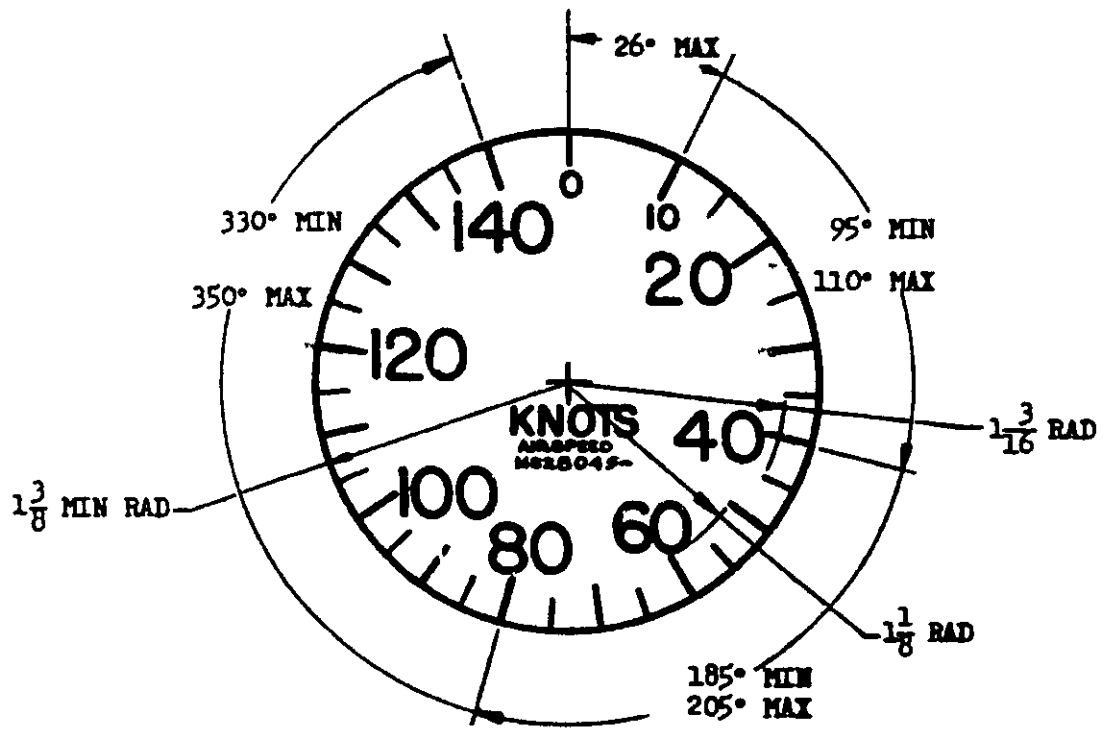
3.7 Markings.

3.7.1 *Case connections.* The back of the case adjacent to the applicable connection shall be marked with the letter "P" for the pressure connection and with the letter "S" for the static connection The following shall also be marked on the back of the case

"CAUTION: NO NOT BLOW IN TUBES"

3 7 2 *Dial* Dial markings shall be durable and the size, material, and finish shall conform to figure 1

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/'' GRADUATIONS ARE PARTIAL RADIAL LINES.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: FRACTIONS ±1/64.

Marking	Height or length ± .010	Width of line or graduation ± .005	Material or finish
Numerals 20, 40, 60, 80, 100, 120, 140	312	.031	Fluorescent luminescent material for Air Force, lusterless white paint for Navy
Graduations corresponding to 10, 20, 30, 40 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150 knot readings	As shown on Figure	.031	
Graduations corresponding to 15, 25, 35, 45, 55, 65, 75, 85, 95, 105, 115, 125, 135, 145 knot readings	As shown on Figure	.016	
Numerals 0 & 10	156	016	
Graduation corresponding to 0	125	.016	
Lettering "Knots"	188	.016	
Lettering "MS28045—"	047	.016	Lusterless black
Lettering "Airspeed"	047	.016	
Background of dial	—	—	

FIGURE 1 *Dial Markings*

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3.7.3 *Fluorescent luminescent material.* The fluorescent luminescent material shall conform to Specification MIL-L-25142

3.7.4 *Lusterless material.* The lusterless white material shall be Color No 37875 of Federal Standard No 595 The lusterless black material shall be Color No 37038 of Federal Standard No 595

3.7.5 *Visibility of dial markings.* All graduations, numerals, pointer, and other markings on the dial shall be readable from any point within the frustum of a cone, the side of which makes an angle of not less than 30 degrees with a perpendicular to the dial and the small diameter of which is the aperture of the dial.

3.8 **Identification of product.** A nameplate, as specified in Specification MIL-I-5415, shall be marked with the following information:

INDICATOR, INDICATED AIRSPEED, 10-150 KNOTS
MS Part No.
Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.[1]
Contract or order No
Manufacturer's name or trademark
US

[1] For instruments furnished on Navy contracts, the space following the words "Serial No." shall be left blank.

3.9 **Workmanship.** Workmanship shall be in accordance with Specification MIL-I-5415

4. QUALITY ASSURANCE PROVISIONS

4.1 **Inspection responsibility.** The supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Samples of the indicators shall be selected in accordance with Specification MIL-I-5415 The inspection and testing of the indicators shall be in accordance with Specification MIL-I-5415 except that the test points, allowable error, and pressure calibration values in inches of water shall be as shown in tables I and II of this specification

5. PREPARATION FOR DELIVERY.

5.1 **Preservation and packaging.**

5.1.1 *Levels A and B.* Preservation shall conform to method IA of Specification MIL-P-116 Packaging shall be in accordance with level A or B, as specified, (see 6.3) of Specification MIL-P-7936 Each unit package shall be overboxed in containers conforming to type I or II, class 1 or 2 of Specification PPP-B-636

5.1.2 *Level C.* The requirements of Specification MIL-P-7936 for Level C shall be applicable

5.2 **Packing.**

5.2.1 *Levels A and B.* The requirements of Specification MIL-P-7936 for these levels apply for overseas and domestic shipment, as applicable

5.2.2 *Level C.* Level C packing shall conform to level C requirements of Specification MIL-P-7936

5.3 **Marking** Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129

5.3.1 *Precautionary marking.* The following precautionary marking shall apply on

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two opposite sides of each interior package and shipping container whenever practicable depending on the size of the carton

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

TABLE I *Test values*

Tests	Tolerances (values in knots unless otherwise specified)
Damping	
Time required from full-scale deflection to 15 knots	1.0 ± 0.2 second
Low temperature exposure and operation	2.0
High temperature operation	2.0
Aging	1.5
Vibration failure	
Test differential pressure equivalent to 120 knots	2.0
Friction	2.0
Position error	2.0
Pointer zero position	
Indication to 0	1/16 inch
Vibration error	
Test differential pressure equivalent to 80 and 120 knots	2.0
Overpressure	
Pitot pressure connection 80 inches of water	
Static pressure connection 60 inches of water	2.0
Seasoning	1.5

TABLE II *Scale errors at room temperature*

Speed (knots)	Pressure (inches of water)	Allowable scale errors (knots)
10	0.0653	2.5
15 [1]	0.1469	2.5
20 [1]	0.2611	2.5
30	0.5876	2.5
40 [1]	1.045	2.0
50	1.634	2.0
60 [1]	2.354	2.0
70	3.207	2.0
80 [1]	4.192	2.0
90	5.310	2.0
100 [1]	6.563	2.0
110	7.951	2.0
120 [1]	9.475	2.0
130	11.136	2.0
140 [1]	12.935	2.0
150	14.873	2.5

[1] Test points

6 NOTES

6.1 Intended use The airspeed indicators covered by this specification are intended for use on hovering observation aircraft

6.2 General. The instructions contained in section 6 including provisions for qualification tests of Specification MIL-I-5415 are applicable to and form a part of this specification except as specified herein

6.3 Ordering data Procurement documents should specify the following

(a) Title, number, and date of this specification

(b) Levels of packaging and packing (see 5.1 and 5.2)

6.4 Qualification With respect to products requiring qualification, awards will be made only for such products as have prior to the time set for opening of bids been tested and approved for inclusion in the applicable Qualified Products List whether or not such

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products have actually been so listed by that date The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be warded contracts or orders for the products covered by this specification The activity responsible for the Qualified Products List is the Bureau of Naval Weapons, Navy Department, Washington 25, D C , and information pertaining to qualification of products may be obtained from that activity

Notice When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto

Custodians
Army—TC
Navy—Wep
Air Force—AFSC

Preparing activity
Navy—Wep

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119 R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner and send to preparing activity *(as indicated on reverse hereof)*

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1426 REPLACES NAVSHIPS FORM 4863 WHICH IS OBSOLETE

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